

ATRIK SAMANTA

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EDUCATION

B.Tech in Computer Science and Engineering (AI & ML) Vellore Institute of Technology (VIT), Vellore CGPA: 7.9	2023 - 2027
Class XII (CBSE) Hem Sheela Model School	2023
Class X (CBSE) Hem Sheela Model School	2021

SKILLS AND INTERESTS

Languages	Python, C, C++, Java, R, SQL, HTML, JavaScript, CSS
Developer Tools	Git, GitHub, VS Code, Jupyter Notebook, Google Colab
Frameworks & Cloud	OpenCV, NumPy, Pandas, MySQL, AWS
Soft Skills	Problem Solving, Teamwork, Communication, Time Management, Adaptability
Spoken Languages	English, Hindi, Bengali, German
Areas of Interest	Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, NLP

PROJECTS

StructuralEye: Crack Detection System <i>Personal Project</i>	March 2026 - April 2026
- Developed an automated crack detection system using image processing (Canny edge detection, CLAHE, morphological operations). - Extracted crack features such as area, length, and width and computed a risk score (0–100) for severity classification. - Classified cracks into LOW, MEDIUM, HIGH and visualized crack growth over time using an interactive dashboard.	
Smart Waste Segregation System <i>Personal Project</i>	September 2025 - November 2025
- Developed an intelligent waste classification system using transfer learning with MobileNetV2 to classify waste into recyclable and non-recyclable categories. - Preprocessed images (resizing to 224x224, normalization, augmentation) and trained on merged datasets (Kaggle + TrashBox). - Achieved 92–95% validation accuracy using CNN with Adam optimizer. - Enabled real-time prediction and integration with smart bins for automated waste segregation.	
AI-Based Fake News Detection System <i>Personal Project</i>	December 2025 - January 2026
- Developed a machine learning model to classify news articles as real or fake using NLP techniques and text preprocessing. - Applied TF-IDF vectorization for feature extraction and trained multiple classifiers including Logistic Regression and Naive Bayes. - Evaluated model performance using accuracy, precision, recall, and F1-score to compare different algorithms. - Achieved high classification accuracy and improved detection reliability through data preprocessing and model tuning.	

Intrusion Detection System for IoT Networks
Personal Project

July 2026 - Present

- Developing an AI-driven Intrusion Detection System (IDS) to actively detect, classify, and mitigate cyber threats and anomalies across Internet of Things (IoT) networks.
- Preprocessing and analyzing high-dimensional network traffic datasets using machine learning and deep learning algorithms to identify malicious patterns and Denial-of-Service (DoS) attacks.
- Optimizing feature engineering and model architecture to achieve high classification accuracy and low latency, ensuring suitability for resource-constrained IoT environments.

AI-Based Logistics Network Optimizer

June 2026 – July 2026

Python, Google OR-Tools, SQLite, Streamlit, Plotly, NetworkX, Pandas, Machine Learning

- Engineered a supply chain optimization system leveraging Operations Research and Linear Programming (OR-Tools) to optimize transportation networks under capacity and demand constraints.
- Built an interactive analytics dashboard for shipment planning, warehouse utilization monitoring, and logistics network visualization, enabling data-driven decision making.
- Developed a relational SQLite backend to store logistics data, shipment plans, and optimization results with SQL-based data retrieval.
- Enhanced the optimization pipeline with machine learning-based demand forecasting (Random Forest/XGBoost), enabling predictive logistics planning.

Vehicle Routing Problem with Time Windows (VRPTW)

July 2026 – August 2026

Python, Google OR-Tools, SQLite, Streamlit, Folium, Plotly, Pandas, OSRM

- Developed a Vehicle Routing Problem with Time Windows (VRPTW) optimization system using Google OR-Tools' Constraint Programming Routing Engine to generate optimal delivery routes while satisfying vehicle capacity, customer time-window, and route duration constraints.
- Built an interactive Streamlit dashboard for uploading delivery datasets, configuring fleet parameters, visualizing optimized vehicle routes on Folium maps, and analyzing delivery schedules using Plotly Gantt Charts.
- Designed a SQLite database to manage depots, vehicles, customers, route assignments, and delivery schedules with SQL-based storage and retrieval of optimized routing solutions.
- Integrated OSRM-based road network distance and travel time matrices with disjunction-based infeasibility handling to optimize real-world transportation networks and improve route planning efficiency.

CERTIFICATIONS

IBM Advanced Generative AI Certification
IBM Career Education

May 2026 - July 2026

- Acquired comprehensive knowledge of Generative AI architectures, including Foundation Models, Transformers, LLMs, GANs, VAEs, and Diffusion Models.
- Applied advanced prompt engineering methodologies such as Zero-shot, Few-shot, Chain-of-Thought (CoT), Tree-of-Thought (ToT), and Persona prompting.
- Gained hands-on experience utilizing IBM watsonx.ai, IBM Granite, Hugging Face, and GitHub Copilot for LLM evaluation and Responsible AI implementation.

POSITIONS OF RESPONSIBILITY

Events Head
The Otaku Club

2024 - 2025

- Formulated, structured, and executed various club events and interactive student activities.
- Led cross-functional student teams to coordinate logistics, marketing, and operations for club-hosted events.

ACHIEVEMENTS

- Successfully built and deployed multiple end-to-end AI/ML projects focusing on Computer Vision and Natural Language Processing.
- Elected and served as the **Events Head** of The Otaku Club for the academic year 2024 – 2025.